



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2011
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #2

Mathematics

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- Each question is to be returned in a separate bundle.
- All necessary working should be shown in every question.

Total Marks – 70

- Attempt questions 1 – 3
- All questions are not of equal value.
- Unless otherwise directed give your answers in simplest exact form.

Examiner: *A.M.Gainford*

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, x > 0$

START A NEW BOOKLET

Question 1. (22 marks)

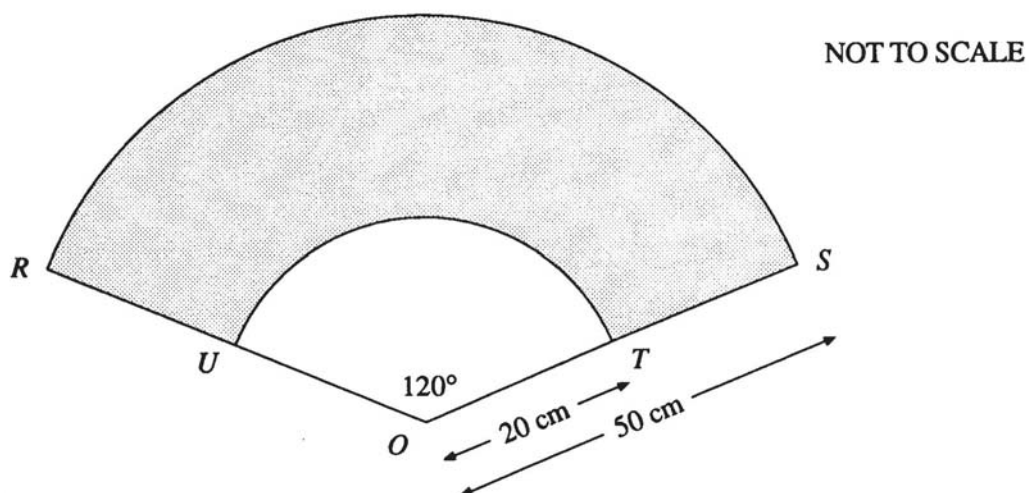
	Marks
(a) Evaluate $\left(\frac{49}{16}\right)^{-\frac{3}{2}}$, as a common fraction in simplest form.	2
(b) (i) Express 1.352 radians in degrees, correct to the nearest minute.	2
(ii) Find $\sin 5$, correct to four significant figures.	
(c) Differentiate	8
(i) $3x^2 - 5x + 7$	
(ii) $\frac{3}{x^4}$	
(iii) $(x^2 - 1)^5$	
(iv) $x^2(1 - x)^4$	
(d) Consider the function $y = 3\sin 2x$.	4
(i) State the amplitude and period of the function.	
(ii) Sketch the graph of the function in the domain $0 \leq x \leq 2\pi$	
(e) (i) Find $\int (3x^2 + 4x - 7)dx$	6
(ii) Evaluate $\int_0^3 (2x^2 + x)dx$	
(iii) Find $\int \frac{1 - x^3}{x^2} dx$	

Question 2 (25 marks)

Marks
4

- (a) (i) Find $\log_3 81$.
- (ii) Given that $\log_4 9 = 1.585$, correct to 3 decimal places, find $\log_4 144$.

(b) 4



A car windscreen wiper sweeps out the shape $RSTU$, where RS and UT are arcs of circles centre O . Measurements are as shown in the figure.

- (i) Calculate the perimeter of $RSTU$.
- (ii) Calculate the area $RSTU$.

(c) 4

x	1	2	3	4	5
$f(x)$					

- (i) Copy and complete the table for $f(x) = \frac{3\sqrt{x}}{x+1}$, correct to 4 decimal places.
- (ii) Using Simpson's Rule with the above function values find an estimate for $\int_1^5 \frac{3\sqrt{x}}{x+1} dx$, correct to 4 significant figures.

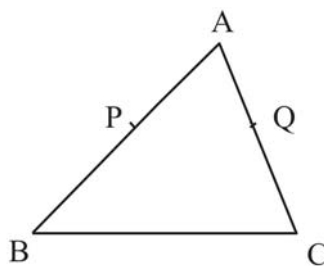
(d) Find that part of the domain for which $f(x) = 2x^3 + 3x^2 - 36x + 1$ is a decreasing function. 2

(e) Given three points $A(0, \sqrt{3})$, $B(3, 0)$, $C(2, -\sqrt{3})$: 5

- (i) Draw a diagram to represent this situation.
- (ii) Show that AB and BC meet at right angles.
- (iii) If D is the point $(1, 0)$, show that A , B and C lie on a circle with centre D .

(f) In the adjoining figure P and Q are the midpoints of AB and AC respectively. 3

Prove that $PQ \parallel BC$, and that PQ is half the length of BC .



(g) The area under the curve $y = 1 - x^2$ from $x = 0$ to $x = 1$ is rotated about the x -axis. 3
Find the volume of the solid of revolution generated.

Question 3 (23 Marks)

Marks
4

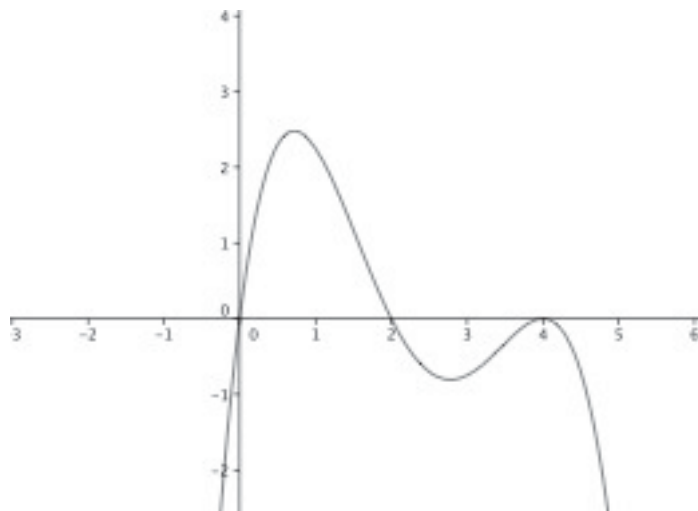
- (a) Two identical urns each contain a number of numbered pool balls. Urn A contains three balls numbered 1, 2, 3, whereas Urn B contains five balls numbered 1, 3, 5, 7, 9.

A ball is drawn at random from each urn.

- (i) What is the probability that both balls have the same number?
- (ii) What is the probability that at least one ball is a 3?

- (b) The diagram shows the graph of a certain derivative, $y = f'(x)$.

3



- (i) Copy this diagram to your Writing Booklet.
- (ii) On the *same* set of axes, draw a sketch of a possible $f(x)$.

- (c) Consider the curves $y = x^2 - 4$ and $y = 2 - x^2$.

4

- (i) Sketch the graphs of the curves on the same axes, and state the x-values of the points of intersection.
- (ii) Hence find the area bounded by the two curves, between their points of intersection.

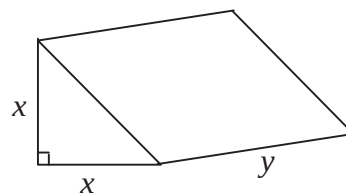
(d) Given the curve $y = x^3 - 6x^2 - 15x$ where $-3 \leq x \leq 9$.

6

- (i) Find any stationary points, and points of inflexion.
- (ii) Sketch the curve, showing its principal features.
- (iii) State the maximum and minimum values of y in the domain.

(e) An isosceles right-triangle based prism, with dimensions x cm and y cm (as shown), is to have a volume of 1000 cm^3 .

6



- (i) Write equations for the volume (V) and surface area (S) of the figure.
- (ii) Show that the surface area $S = x^2 + \frac{2000(2 + \sqrt{2})}{x}$.
- (iii) Find the value of x (correct to one decimal place) so that the surface area is a minimum.

This is the end of the paper.

START A NEW BOOKLET

Question 1. (22 marks)

$$\left(\frac{16}{49}\right)^{\frac{3}{2}} = \frac{4^3}{7^3} = \frac{64}{343} \quad \text{Marks } 2$$

(a) Evaluate $\left(\frac{49}{16}\right)^{\frac{3}{2}}$, as a common fraction in simplest form.

(b) (i) Express 1.352 radians in degrees, correct to the nearest minute.

$$\pi^c = 180^\circ$$

$$1^c = \frac{180}{\pi}$$

$$1.352 \times \frac{180}{\pi} = 77^\circ 28'$$

(ii) Find $\sin 5$, correct to four significant figures.

$$-0.9589 \quad 1$$

(c) Differentiate

(i) $\frac{d}{dx}(3x^2 - 5x + 7) = 6x - 5 \quad 1$

(ii) $\frac{d}{dx}\left(\frac{3}{x^4}\right) = \frac{d}{dx}(3x^{-4}) = -12x^{-5} = -\frac{12}{x^5} \quad 2$

(iii) $\frac{d}{dx}(x^2 - 1)^5 = 5(x^2 - 1)^4 \times 2x = 10x(x^2 - 1)^4 \quad 2$

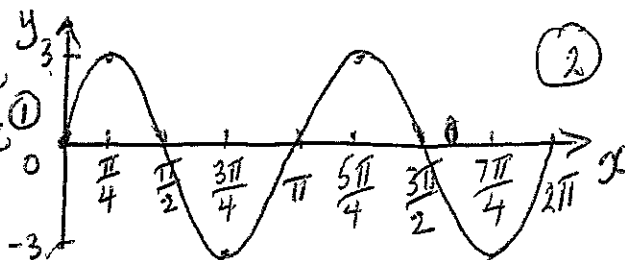
(iv) $\frac{d}{dx}x^2(1-x)^4 = x^2 \times 4(1-x)^3 \times -1 + (1-x)^4 \times 2x$
 $= 2x(1-x)^3[-2x + (1-x)] = 2x(1-x)^3[-3x+1] \quad 3$

(d) Consider the function $y = 3\sin 2x$.

amplitude $3 \quad 1$
 period $= \frac{2\pi}{2} = \pi \quad 4$

(i) State the amplitude and period of the function.

(ii) Sketch the graph of the function in the domain $0 \leq x \leq 2\pi$



(e) (i) Find $\int(3x^2 + 4x - 7)dx$

$$= \frac{3x^3}{3} + \frac{4x^2}{2} - 7x + C = x^3 + 2x^2 - 7x + C \quad 1$$

(ii) Evaluate $\int_0^3(2x^2 + x)dx$

(iii) Find $\int \frac{1-x^3}{x^2} dx$

$$= \int \frac{1}{x^2} - \frac{x^3}{x^2} dx$$

$$= \int x^{-2} - x dx$$

$$= \frac{x^{-1}}{-1} - \frac{x^2}{2} + C$$

$$= -\frac{1}{x} - \frac{x^2}{2} + C \quad 3$$

$$\left[\frac{2x^3}{3} + \frac{x^2}{2} \right]_0^3$$

$$= \left(\frac{2 \times 27}{3} + \frac{9}{2} \right) - (0+0)$$

$$= 18 + 4\frac{1}{2}$$

$$= 22\frac{1}{2} \quad 2$$

Question 2

a.i. $\log_3 81 = 4$ (1)

ii $\log_4 144 = \log_4 9 + \log_4 16$ (1)
 $= 1.585 + \log_4 4^2$
 $= 1.585 + 2\log_4 4$ (2)
 $= 1.585 + 2$
 $= 3.585$ (3)

b. $120^\circ = 2\pi/3$

i. $P = 2(50-20) + 20 \times 2\pi/3 + 50 \times 2\pi/3$
 $= 206.61 \text{ cm}$ (2dp) (2)

ii $A = (\frac{1}{2} \times 50^2 \times 2\pi/3) - (\frac{1}{2} \times 20^2 \times 2\pi/3)$
 $= 2199.11 \text{ cm}^2$ (2dp) (2)

c.

i.	x	1	2	3	4	5
	$f(x)$	1.5	1.442	1.2990	1.2	1.1180

(1)

ii $h = \frac{5-1}{4} = \frac{4}{4} = 1$

$$\int_1^5 \frac{3\sqrt{x}}{x+1} dx$$

$$\approx \frac{1}{3} \left[1.5 + 1.1180 + 4(1.442 + 1.2) + 2(1.2990) \right]$$

$= 5.224$ (4 sig fig) (3)

d. decreasing $\therefore f'(x) < 0$

$f'(x) = 6x^2 + 6x - 36$ (1/2)

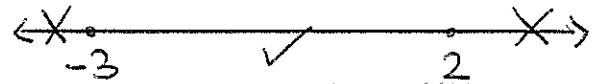
St. at $f'(x) = 0$

$6x^2 + 6x - 36 = 0$

$x^2 + x - 6 = 0$

$(x+3)(x-2) = 0$

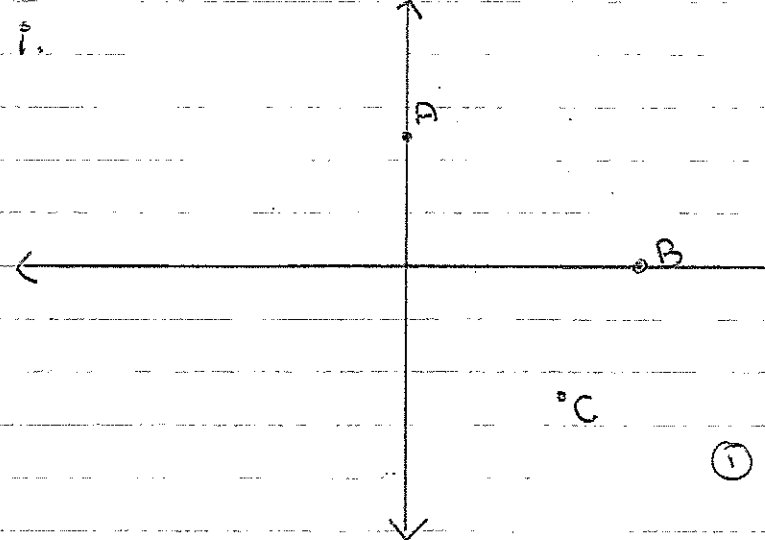
$\therefore x = -3, x = 2$



$f'(-4) = \frac{36}{36}$ $f'(0) = \frac{-36}{-36}$ $f'(3) = \frac{36}{36}$

$\therefore -3 < x < 2$ (2)

e



(1)

i. M of AB = $\frac{\sqrt{3}-0}{0-3}$

$= \frac{-\sqrt{3}}{3}$

$= -\frac{\sqrt{3}}{3}$

M of BC = $\frac{0+\sqrt{3}}{3-2}$

$= \frac{\sqrt{3}}{1}$

$= \sqrt{3}$

$M_1 \times M_2 = -\frac{\sqrt{3}}{3} \times \sqrt{3}$

$= -\frac{3}{3}$

$= -1$

$\therefore AB \perp BC$

(2)

$$\text{iii. } \text{d of } BD = \sqrt{(3-1)^2 + (0-0)^2} \\ = \sqrt{2^2} \\ = 2$$

$\therefore$ centre $(1, 0)$ radius 2

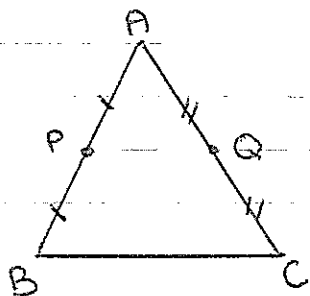
circle eqn:
 $(x-1)^2 + y^2 = 4$

Check A: $(0-1)^2 + (\sqrt{3})^2$
 $= 1 + 3$
 $= 4$

Check B: $(3-1)^2 + 0^2$
 $= 2^2$
 $= 4$

Check C: $(2-1)^2 + (-\sqrt{3})^2$
 $= 1 + 3$
 $= 4$

$\therefore$ A, B & C lie on a circle with centre D. ②



$\triangle APQ \sim \triangle ABC$ as $\angle A$ is common & $\frac{AP}{AB} = \frac{1}{2} = \frac{AQ}{AC}$

"Two triangles are similar if there is an equal angle & the sides making this angle are in the same ratio."

$\therefore \angle APQ = \angle ABC$
 & $\angle AQP = \angle ACB$
 (Corresponding $\angle$'s in $\parallel \Delta$'s)

$\therefore PQ \parallel BC$. ②

$\frac{PQ}{BC} = \frac{AP}{AB}$ (sides in same ratio)

$\frac{PQ}{BC} = \frac{1}{2}$

$PQ = \frac{1}{2} BC$. ①

g. $V = \pi \int_0^1 (1-x^2)^2 dx$ ①

$= \pi \int_0^1 (1 - 2x^2 + x^4) dx$

$= \pi \left[x - \frac{2x^3}{3} + \frac{x^5}{5} \right]_0^1$ ②

$= \pi \left[\left(1 - \frac{2}{3} + \frac{1}{5} \right) - 0 \right]$

$= \frac{8\pi}{15}$. ③

Solns 2 Unit Yr 12

23

3(a)

1, 2,
3

A

1, 3, 5,
7, 9

B

(i) $P(1, 1 \text{ or } 3, 3)$

$$= \frac{1}{3} \times \frac{1}{5} + \frac{1}{3} \times \frac{1}{5}$$

$$= \frac{1}{15} + \frac{1}{15}$$

$$= \frac{2}{15}$$

(2)

(ii) $P(\text{at least 1 ball is a 3})$

$$= P(3, \bar{3} \text{ or } \bar{3}, 3) \text{ or } 3, 3 \text{ OR } 1 - P(\bar{3}, \bar{3})$$

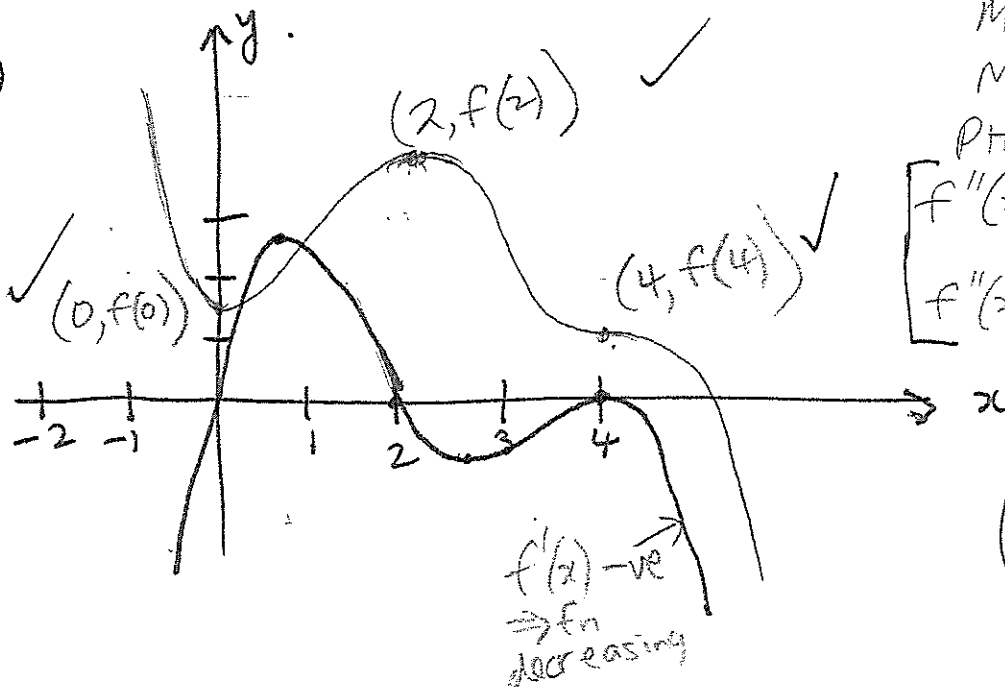
$$= \frac{1}{3} \times \frac{4}{5} + \frac{2}{3} \times \frac{1}{5} + \frac{1}{3} \times \frac{1}{5}$$

$$= \frac{4}{15} + \frac{2}{15} + \frac{1}{15}$$

$$= \frac{7}{15}$$

(2)

b)



Min TP at $x=0$

Max TP at $x=2$

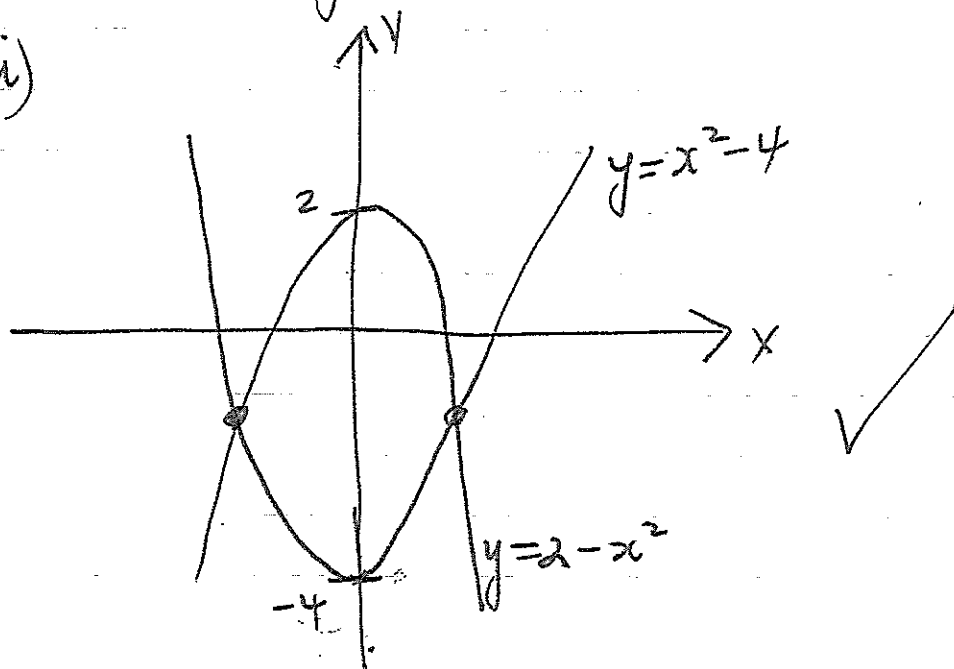
PHI at $x=4$

$$\begin{aligned} f''(x) &= 0 \text{ at } x \doteq 1.8 \\ &\Rightarrow \text{change in concavity} \\ f''(x) &= 0 \text{ at } x \doteq 2.9 \\ &\Rightarrow \text{change in concavity} \end{aligned}$$

(3)

3(c) $y = x^2 - 4$ ①
 $y = 2 - x^2$ ②

(i)



Points of intersection - solve simultaneously

$$\Rightarrow x^2 - 4 = 2 - x^2$$

$$2x^2 = 6$$

$$x^2 = 3$$

$$x = \pm\sqrt{3}$$



②

(ii) $A = \int_{-\sqrt{3}}^{\sqrt{3}} (2 - x^2) - (x^2 - 4) dx$ ✓

$$= \int_{-\sqrt{3}}^{\sqrt{3}} (-2x^2 + 6) dx$$

$$= \left[-\frac{2x^3}{3} + 6x \right]_{-\sqrt{3}}^{\sqrt{3}} = \left(\frac{-2(\sqrt{3})^3}{3} + 6\sqrt{3} \right) - \left(\frac{-2(-\sqrt{3})^3}{3} + 6(-\sqrt{3}) \right)$$

$$= -\frac{2\sqrt{3}^3}{3} + 6\sqrt{3} - \frac{2\sqrt{3}^3}{3} + 6\sqrt{3}$$

$$A = -4\sqrt{3} + 12\sqrt{3} = 8\sqrt{3} \text{ sq. units.}$$

②

(d) $y = x^3 - 6x^2 - 15x$, $-3 \leq x \leq 9$

(i) $y' = 3x^2 - 12x - 15$

$y'' = 6x - 12$

For t.p's $y' = 0 \Rightarrow 3(x^2 - 4x - 5) = 0$ ✓
 $3(x-5)(x+1) = 0$
 $\Rightarrow \underline{x = 5 \text{ or } -1}$

When $x = 5$, $y = -100 \rightarrow (5, -100)$ ✓
 $x = -1$, $y = 8 \rightarrow (-1, 8)$

Type of St. Point

$y''(5) = 30 - 12 > 0 \Rightarrow \underline{\text{min at } (5, -100)}$

$y''(-1) = -6 - 12 < 0 \Rightarrow \underline{\text{max at } (-1, 8)}$

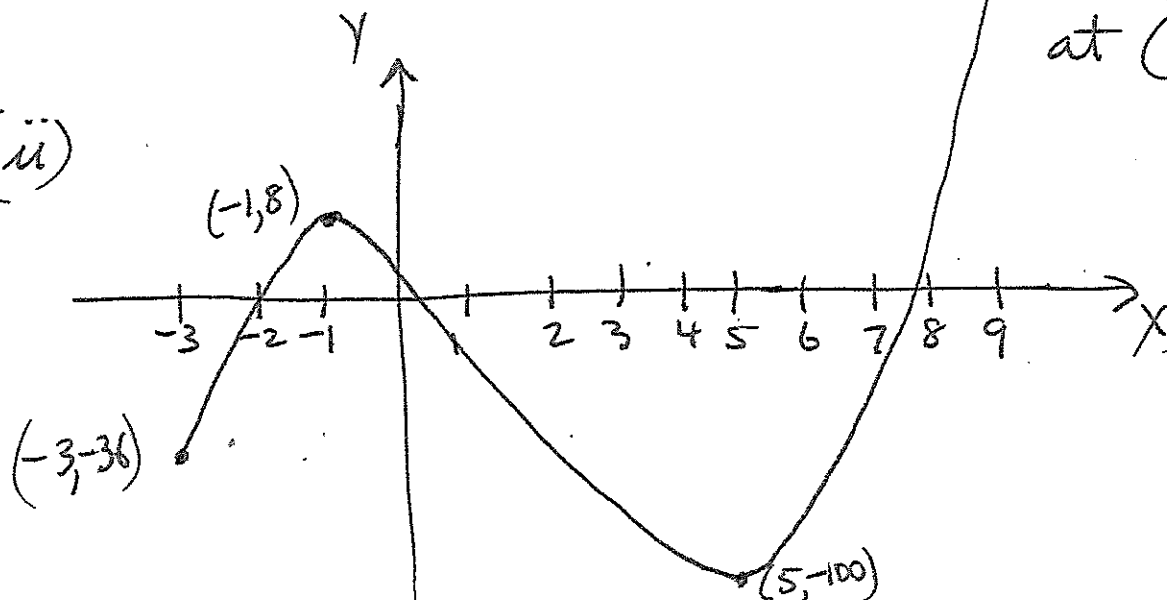
Points of Inflection When $y'' = 0$

$\Rightarrow 6x - 12 = 0$

$\underline{\underline{x = 2}}$

$(9, 108)$ ✓
 (Change in concavity at $(2, -46)$) ✓

(ii)



1

4

3.(d)(iii) When $x = -3$, $y = -36$
 $x = 9$, $y = 108$] End values

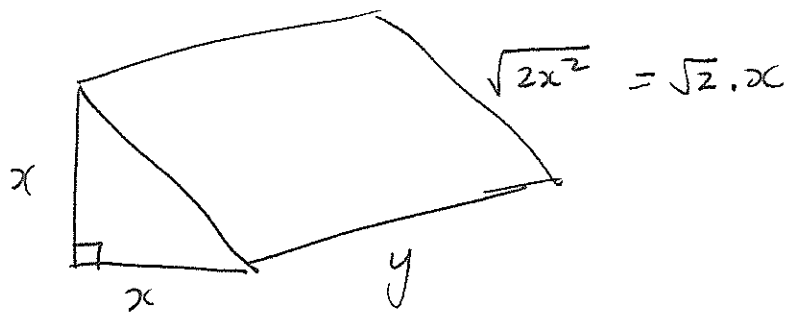
$x = -1$, $y = 8$
 $x = 5$, $y = -100$] Turn. Points

$\therefore$ Minimum y value 108

Max y value = -100

✓ (1)

3(e)



$$(i) \quad V = \frac{x^2}{2} \cdot y \quad (1)$$

$$S = x^2 + \sqrt{2}xy + 2xy$$

$$= x^2 + (2 + \sqrt{2})xy \quad (2)$$

1

$$(ii) \quad \text{Show } S = x^2 + \frac{2000(2 + \sqrt{2})}{x}$$

$$\text{Now } \frac{x^2 y}{2} = 1000$$

$$\Rightarrow y = \frac{2000}{x^2} \quad (3) \checkmark$$

Sub (3) in (2)

$$\Rightarrow S = x^2 + (2 + \sqrt{2})x \cdot \frac{2000}{x^2}$$

$$S = x^2 + \frac{2000(2 + \sqrt{2})}{x} \quad \checkmark$$

2

$$(iii) \quad S' = 2x - \frac{2000(2 + \sqrt{2})}{x^2} \quad \checkmark$$

$$\text{For turn. pts, } S' = 0 \Rightarrow 2x - \frac{2000(2 + \sqrt{2})}{x^2} = 0$$

$$2x^3 = 2000(2 + \sqrt{2})$$

$$x^3 = 1000(2 + \sqrt{2})$$

$$x = 15.0578$$

$$\underline{x = 15.1 \text{ to 1dp}} \quad \checkmark$$

3(e) (iii) (cont)

$$S'' = 2 + \frac{4000(2+\sqrt{2})}{x^3}$$

$$S''(15.1) = 2 + \frac{4000(2+\sqrt{2})}{(15.1)^3} > 0 \quad \checkmark$$

$\Rightarrow$ Min. at $x = 15.1$

ie. SA. is a minimum when $x = 15.1$

(3)